1- CHIFFREMENT SYMÉTRIQUE

Créer le fichier

```
┌──(khouloud㉿kali)-[~]
└─$ touch fichier_khouloud
```

Chiffrement RC4

```
┌──(khouloud㉿kali)-[~]
└─$ openssl enc -rc4 -in fichier_khouloud -out fichier_chiff_rc4

enter RC4 encryption password:
Verifying - enter RC4 encryption password:
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
```

Vérifier que le résultat est illisible:

```
┌──(khouloud㉿kali)-[~]
└─$ cat fichier_chiff_rc4

Salted__x◆◆
           ◆h◆
```

Déchiffrement RC4

```
┌──(khouloud㉿kali)-[~]
└─$ openssl enc -rc4 -d -in fichier_chiff_rc4 -out fichier_dechiff_rc4

enter RC4 decryption password:
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
```

Vérification :

```
┌──(khouloud㉿kali)-[~]
└─$ diff fichier_khouloud fichier_dechiff_rc4


┌──(khouloud㉿kali)-[~]
└─$
```

rien ne s'affiche → les deux fichiers sont identiques → **décryptage réussi**.

Chiffrement DES (clé explicite):

```
┌──(khouloud㉿kali)-[~]
└─$ openssl enc -des -in fichier_khouloud -out fichier_chiff_des -k 0123456789ABCDEF

*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
```

Déchiffrement DES

```
┌──(khouloud㉿kali)-[~]
└─$ openssl enc -des -d -in fichier_chiff_des -out fichier_dechiff_des -k 0123456789ABCDEF

*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
```

Vérification :

```
┌──(khouloud㉿kali)-[~]
└─$ diff fichier_khouloud fichier_dechiff_des


┌──(khouloud㉿kali)-[~]
└─$
```

2- CHIFFREMENT ASYMÉTRIQUE RSA

Génération de la clé privée RSA (1024 bits)

```
┌──(khouloud㉿kali)-[~]
└─$ openssl genrsa -out rsakey.pem 1024


┌──(khouloud㉿kali)-[~]
└─$
```

Afficher

```
┌──(khouloud㉿kali)-[~]
└─$ cat rsakey.pem

-----BEGIN PRIVATE KEY-----
MIICeAIBADANBgkqhkiG9w0BAQEFAASCAmIwggJeAgEAAoGBAK1DveD6awDuM0Zr
Q0sTzZKWyWEZWeSWJ3Aec31n8waQv1DQa7c2/tR6ZBIksBU+xlXrpggq6gtMyq6u
ann2SJbsHcX0LANtaIE/RQ9rVRjRBmPW9Vw+QHGcWv+PBz04UmIhhjsDFSxxj/JR
zcEA7LJQT1m14jttwbrI+B1oMyfZAgMBAAECgYEAqyb5kB2OSdMC75LRXH5pWV59
+t6x/62bs5NA/iYK2PdvJQNdBBTdUUK9S3bWQYZLIWCwF+1cWGE47aj5IRT5ygvA
LeJQxPu2Ek728I3wB1Gn5jWbfgNI5G36gLS4e1kqExBxcExdM+tcUHR55LCZHBgT
6i7BLhghXXUwIfzkOMkCQQDVZgp2FY+/TplVKjF4i2xK75tf9IFUC5+FGESetbmM
9BqXZE4oYsfSZ+CyUjMXOfnPeXakWC8tUx8Y8Zo2sPd7AkEAz9qclT35lQSprQ9U
d1pSKLiEwnJS/8EVmtf3c7nzJTar2EkKCd3uDPwxCJrkbugoYmcDMQDDdjjScCPH
PpzTuwJBAL1KP1tRcun6eW6e97T6z3uFIOizxQhdKwnvZeb2QXnziabitaWDQca0
mA71dRZUOfA8yO2sv2RQCBSjMuzyewMCQQCSLUJ1yvIdlPvnU6KplYG2VZ44ScKJ
TzdsgHnsj1eqUjqkjylaKxPhWfL6BESmourT2XJrIghUuj101BSpug1FAkBVCHLv
kvvsQny3T+wqIvOIXQeofrBqrpJ8U2bhAwdW9hIelm/WJGOsvgkbtt4LwfTweZI9
R5O+9Zmr5w8i5R0j
-----END PRIVATE KEY-----
```

Afficher en texte clair + hex :

```
┌──(khouloud㉿kali)-[~]
└─$ openssl rsa -in rsakey.pem -text -noout

Private-Key: (1024 bit, 2 primes)
modulus:
    00:ad:43:bd:e0:fa:6b:00:ee:33:46:6b:43:4b:13:
    cd:92:96:c9:61:19:59:e4:96:27:70:1e:73:7d:67:
    f3:06:90:bf:50:d0:6b:b7:36:fe:d4:7a:64:12:24:
    b0:15:3e:c6:55:eb:a6:08:2a:ea:0b:4c:ca:ae:ae:
    6a:79:f6:48:96:ec:1d:c5:f4:2c:03:6d:68:81:3f:
    45:0f:6b:55:18:d1:06:63:d6:f5:5c:3e:40:71:9c:
    5a:ff:8f:07:3d:38:52:62:21:86:3b:03:15:2c:71:
    8f:f2:51:cd:c1:00:ec:b2:50:4f:59:b5:e2:3b:6d:
    c1:ba:c8:f8:1d:68:33:27:d9
publicExponent: 65537 (0×10001)
privateExponent:
    00:ab:26:f9:90:1d:8e:49:d3:02:ef:92:d1:5c:7e:
    69:59:5e:7d:fa:de:b1:ff:ad:9b:b3:93:40:fe:26:
    0a:d8:f7:6f:25:03:5d:04:14:dd:51:42:bd:4b:76:
    d6:41:86:4b:21:60:b0:17:ed:5c:58:61:38:ed:a8:
    f9:21:14:f9:ca:0b:c0:2d:e2:50:c4:fb:b6:12:4e:
    f6:f0:8d:f0:07:51:a7:e6:35:9b:7e:03:48:e4:6d:
    fa:80:b4:b8:7b:59:2a:13:10:71:70:4c:5d:33:eb:
    5c:50:74:79:e4:b0:99:1c:18:13:ea:2e:c1:2e:18:
    21:5d:75:30:21:fc:e4:38:c9
prime1:
    00:d5:66:0a:76:15:8f:bf:4e:99:55:2a:31:78:8b:
    6c:4a:ef:9b:5f:f4:81:54:0b:9f:85:18:44:9e:b5:
    b9:8c:f4:1a:97:64:4e:28:62:c7:d2:67:e0:b2:52:
    33:17:39:f9:cf:79:76:a4:58:2f:2d:53:1f:18:f1:
    9a:36:b0:f7:7b
prime2:
    00:cf:da:9c:95:3d:f9:95:04:a9:ad:0f:54:77:5a:
    52:28:b8:84:c2:72:52:ff:c1:15:9a:d7:f7:73:b9:
    f3:25:36:ab:d8:49:0a:09:dd:ee:0c:fc:31:08:9a:
    e4:6e:e8:28:62:67:03:31:00:c3:76:38:d2:70:23:
    c7:3e:9c:d3:bb
exponent1:
    00:bd:4a:3f:5b:51:72:e9:fa:79:6e:9e:f7:b4:fa:
    cf:7b:85:20:e8:b3:c5:08:5d:2b:09:ef:65:e6:f6:
    41:79:f3:89:a6:e2:b5:a5:83:41:c6:b4:98:0e:f5:
    75:16:54:39:f0:3c:c8:ed:ac:bf:64:50:08:14:a3:
    32:ec:f2:7b:03
```

```
exponent2:
    00:92:2d:42:75:ca:f2:1d:94:fb:e7:53:a2:a9:95:
    81:b6:55:9e:38:49:c2:89:4f:37:6c:80:79:ec:8f:
    57:aa:52:3a:a4:8f:29:5a:2b:13:e1:59:f2:fa:04:
    44:a6:a2:ea:d3:d9:72:6b:22:08:54:ba:3d:74:d4:
    14:a9:ba:0d:45
coefficient:
    55:08:72:ef:92:fb:ec:42:7c:b7:4f:ec:2a:22:f3:
    88:5d:07:a8:7e:b0:6a:ae:92:7c:53:66:e1:03:07:
    56:f6:12:1e:96:6f:d6:24:63:ac:be:09:1b:b6:de:
    0b:c1:f4:f0:79:92:3d:47:93:be:f5:99:ab:e7:0f:
    22:e5:1d:23

┌──(khouloud㉿kali)-[~]
└─$
```

Extraction clé publique

```
┌──(khouloud㉿kali)-[~]
└─$ openssl rsa -in rsakey.pem -pubout -out rsapubkey.pem

writing RSA key

┌──(khouloud㉿kali)-[~]
└─$
```

Chiffrer la clé privée avec AES256

```
┌──(khouloud㉿kali)-[~]
└─$ openssl enc -aes256 -in rsakey.pem -out rsakeyencaes.pem

enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

┌──(khouloud㉿kali)-[~]
└─$
```

Chiffrement du fichier_khouloud avec RSA

```
┌──(khouloud㉿kali)-[~]
└─$ openssl pkeyutl -pubin -inkey rsapubkey.pem -in fichier_khouloud -encrypt -out fichier_khouloud.rsaenc

┌──(khouloud㉿kali)-[~]
└─$
```

Déchiffrement du fichier RSA

```
┌──(khouloud㉿kali)-[~]
└─$ openssl pkeyutl -inkey rsakey.pem -in fichier_khouloud.rsaenc -decrypt -out fichier_khouloud.rsadec

┌──(khouloud㉿kali)-[~]
└─$
```

Vérification :

```
┌──(khouloud㉿kali)-[~]
└─$ diff fichier_khouloud fichier_khouloud.rsadec

┌──(khouloud㉿kali)-[~]
└─$
```

3- SIGNATURE NUMÉRIQUE

Empreinte MD5

```
┌──(khouloud㉿kali)-[~]
└─$ openssl dgst -md5 -out fichier_khouloud.md5 fichier_khouloud

┌──(khouloud㉿kali)-[~]
└─$
```

Taille MD5 = **128 bits = 16 octets**

Empreinte SHA1

```
┌──(khouloud㉿kali)-[~]
└─$ openssl dgst -sha1 -out fichier_khouloud.sha1 fichier_khouloud

┌──(khouloud㉿kali)-[~]
└─$
```

Taille SHA1 = **160 bits = 20 octets**

Comparaison : SHA1 donne une empreinte plus longue → plus sécurisée.

Signature du fichier (on signe l'empreinte SHA1)

Clé utilisée = **clé privée (`rsakey.pem`)**

```
┌──(khouloud㉿kali)-[~]
└─$ openssl rsautl -sign -in fichier_khouloud.sha1 -inkey rsakey.pem -out fichier_sig

The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead.
```

Vérification de la signature

Clé utilisée = **clé publique (`rsapubkey.pem`)**.

```
┌──(khouloud㉿kali)-[~]
└─$ openssl rsautl -verify -in fichier_sig -out empreinte2 -pubin -inkey rsapubkey.pem

The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead.
```

Comparer les empreintes

```
┌──(khouloud㉿kali)-[~]
└─$ diff fichier_khouloud.sha1 empreinte2

┌──(khouloud㉿kali)-[~]
└─$
```

rien ne s'affiche → la signature est correcte.

4 - CERTIFICAT NUMÉRIQUE

Clé privée du serveur

```
┌──(khouloud㉿kali)-[~]
└─$ openssl genrsa -out server_cle.pem 1024

┌──(khouloud㉿kali)-[~]
└─$
```

Création d'une requête CSR

```
┌──(khouloud㉿kali)-[~]
└─$ openssl req -new -key server_cle.pem -out server_cert.csr

You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:TN
State or Province Name (full name) [Some-State]:Tunis
Locality Name (eg, city) []:Ariana
Organization Name (eg, company) [Internet Widgits Pty Ltd]:Esprit
Organizational Unit Name (eg, section) []:IT
Common Name (e.g. server FQDN or YOUR name) []:server_khouloud
Email Address []:

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:loody
An optional company name []:

┌──(khouloud㉿kali)-[~]
└─$
```

-> Crée une demande de certificat.

Auto-signature du certificat (valide 1 an)

```
┌──(khouloud㉿kali)-[~]
└─$ openssl req -new -x509 -days 365 -key server_cle.pem -out server_cert.crt

You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:TN
State or Province Name (full name) [Some-State]:Tunis
Locality Name (eg, city) []:Ariana
Organization Name (eg, company) [Internet Widgits Pty Ltd]:Esprit
Organizational Unit Name (eg, section) []:IT
Common Name (e.g. server FQDN or YOUR name) []:server_khouloud
Email Address []:

┌──(khouloud㉿kali)-[~]
└─$
```

-> Crée un certificat auto-signé pour le serveur.

Afficher son contenu :

```
┌──(khouloud㉿kali)-[~]
└─$ openssl x509 -in server_cert.crt -text -noout

Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            3a:ed:64:b4:94:a4:9a:61:f8:0b:d7:dd:4e:ee:38:fb:a8:31:42:2d
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=TN, ST=Tunis, L=Ariana, O=Esprit, OU=IT, CN=server_khouloud
        Validity
            Not Before: Nov 26 15:01:52 2025 GMT
            Not After : Nov 26 15:01:52 2026 GMT
        Subject: C=TN, ST=Tunis, L=Ariana, O=Esprit, OU=IT, CN=server_khouloud
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (1024 bit)
                Modulus:
                    00:c5:bd:ac:2e:48:15:8c:7a:b5:51:b9:a7:0b:93:
                    35:e4:e6:fe:93:d1:ab:c9:05:e0:c8:4b:78:18:88:
                    50:ed:6a:fe:07:e3:51:e6:cc:a4:f0:75:87:40:85:
                    bf:14:57:c9:7c:df:83:75:c8:74:77:9a:9a:82:80:
                    52:c2:25:ff:6a:cf:4f:6a:10:e0:fe:3a:a7:27:4b:
                    79:a1:06:e8:55:7d:6a:15:be:bc:83:60:b5:79:5c:
                    60:a0:26:2d:ce:a1:db:02:b3:8f:36:ef:1e:36:54:
                    c0:e0:f2:13:07:fa:b5:02:99:0f:58:97:34:f0:7e:
                    e5:9e:33:bb:47:5e:0d:88:ef
                Exponent: 65537 (0x10001)
        X509v3 extensions:
            X509v3 Subject Key Identifier:
                F8:FB:11:2A:06:07:CF:9C:44:D7:1F:40:21:E7:10:51:18:3D:CB:85
            X509v3 Authority Key Identifier:
                F8:FB:11:2A:06:07:CF:9C:44:D7:1F:40:21:E7:10:51:18:3D:CB:85
            X509v3 Basic Constraints: critical
                CA:TRUE
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:
        83:67:82:bd:38:49:d8:70:a1:21:38:39:39:7e:5e:90:ae:f3:
        4d:e0:7d:44:b2:b4:92:26:75:3c:db:f7:67:e9:00:e3:67:d2:
        c8:44:3c:cb:0f:97:71:1a:ce:44:05:73:ad:f4:01:0d:a6:7d:
        c9:97:c6:86:35:c0:61:11:c3:83:f0:d0:99:af:ad:8c:a3:50:
        73:6e:0d:12:f0:0a:c0:05:51:17:ab:26:5c:9f:47:a7:03:88:
```

5- SIGNATURE PAR UNE AUTORITÉ DE CERTIFICATION (AC)

Clé privée AC (2048 bits)

```
┌──(khouloud㉿kali)-[~]
└─$ openssl genrsa -out cakey.pem 2048

┌──(khouloud㉿kali)-[~]
└─$
```

Certificat AC auto-signé

```
┌──(khouloud㉿kali)-[~]
└─$ openssl req -new -x509 -days 730 -key cakey.pem -out ca.crt

You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:TN
State or Province Name (full name) [Some-State]:Tunis
Locality Name (eg, city) []:Ariana
Organization Name (eg, company) [Internet Widgits Pty Ltd]:Esprit
Organizational Unit Name (eg, section) []:IT
Common Name (e.g. server FQDN or YOUR name) []:server_khouloud
Email Address []:

┌──(khouloud㉿kali)-[~]
└─$
```

Signature du certificat serveur par l'AC

```
┌──(khouloud㉿kali)-[~]
└─$ openssl x509 -req -in server_cert.csr -out server_signed.crt -CA ca.crt -CAkey cakey.pem -CAcreateserial -CAserial ca.srl

Certificate request self-signature ok
subject=C=TN, ST=Tunis, L=Ariana, O=Esprit, OU=IT, CN=server_khouloud

┌──(khouloud㉿kali)-[~]
└─$
```